NOVA SCHOOL OF SCIENCE & TECHNOLOGY

MEEC/MEMN

2025 / 2026

Course : **Electrónica de Rádio Frequência**

Low Noise Amplifier – Part 1

Goal:

Analysis and design of a LNA with lumped and distributed elements.

http://moodle.fct.unl.pt

Version 1.0
Date: 1st April 2026
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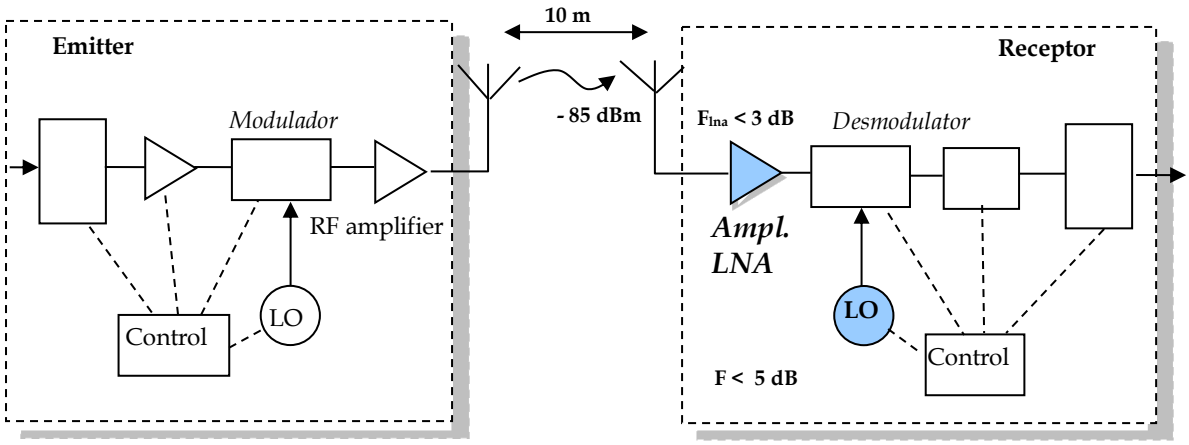
Motivation

Modern telecommunications systems use high-frequencies with high data-rates. In the receiver path the critical block is the LNA, since, it needs to deal with very low amplitude signals. The LNA has stringent specifications in terns of gain, noise, IM3, P^{-1}dB... In this Project it is expected that the students will understand in detail all these aspects designing an LNA for a ISM (industrial, scientific and medical) band.



Specifications

The LNA for ISM band should fulfill the following specifications:

- S11 < - 10 dB (with Matching networks);
- S22 < -10 dB (with Matching networks);
- F < 3 dB;
- Gain = MAG = S21 dB (with Matching networks);
- 50 Ω load and source.
- Characteristic impedance: 50 Ω
- Operation frequency: 3-6 GHz (Bipolar design) (check stability and available gain) please contact professor before start);

Project assignment

The students should form groups of three. All the groups will use the same transistor BPF420 (form Infineon), with different bias point. Therefore, each group will use a different S-parameter file.

Group	Transistor BFP420	Group	Transistor BFP420
T1G1	IC = 4 mA VCE = 3 V	T2G1	IC = 6 mA VCE = 5 V
T1G2	IC = 10 mA VCE = 3 V	T2G2	IC = 9 mA VCE = 5 V
T1G3	IC = 6 mA VCE = 3 V	T2G3	IC = 7 mA VCE = 5 V
T1G4	IC = 13 mA VCE = 3 V	T2G4	IC = 11 mA VCE = 5 V
T1G5	IC = 9 mA VCE = 3 V	T2G5	IC = 4 mA VCE = 5 V
T1G6	IC = 15 mA VCE = 3 V	T2G6	IC = 13 mA VCE = 5 V
T1G7	IC = 17 mA VCE = 3 V	T2G7	IC = 20 mA VCE = 5 V
T1G9	IC = 10 mA VCE = 4 V	T2G9	IC = 20 mA VCE = 4 V

Components

- **Transistors**

Documents about the BFP420 transistor: Datasheet, spice models are available in the moodle.

Design procedure

1- Project:

LNA

a. Design the LNA according with the specifications, with a detailed explanation about:
   i. Design Transistor Bias network;
   ii. Obtain S-parameters with packaging effects;
   iii. Transistor validation for the given bias point;
   iv. Stability and obtain the operation frequency (check with professor)
   v. Design the input and output matching networks.
   vi. Determine the gain and NF.

b. Simulation:
   i. Validate the LNA designs using LTSpice.
   ii. Investigate the possibility of changing the input matching, according with the previous specifications of Gain and NF.

2- Report

a. The report should have:
   i. Detailed justification about the options done during the project.
   ii. Circuit schematics, simulations files, plot, simulation results.

b. Structure:
   i. First page with the group number and students identification.
   ii. Introduction
      1. objective
   iii. Main Body
      1. Design
      2. Simulations
      3. Final circuit
      4. Analysis of Results
   iv. Conclusions

3- Delivery

a. Date: 10-05-2026
b. Place: upload in moodle

Contacts

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